

# Math 220: Homework 2

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February 12, 2026

**Problem 1** (Examples of Topologies). (i) Prove that  $\mathcal{T} = \{(-\infty, x) : x \in \mathbb{R}\} \cup \{\emptyset, \mathbb{R}\}$  is a topology on  $\mathbb{R}$ .

**Proof.** Clearly, we see that  $\emptyset, \mathbb{R}$  by definition of  $\mathcal{T}$ . Let  $A_1, A_2 \in \mathcal{T}$ . If  $A_1 = \emptyset$  and  $A_2 = \emptyset$ , then  $A_1 \cap A_2 = \emptyset \in \mathcal{T}$ . If  $A_1 = \mathbb{R}$  and  $A_2 = \emptyset$ , then again  $A_1 \cap A_2 = \emptyset \in \mathcal{T}$  (same conclusion if we switch roles). If  $A_1 = \mathbb{R}$  and  $A_2 = \mathbb{R}$ , then  $A_1 \cap A_2 = \mathbb{R} \in \mathcal{T}$ . Otherwise,  $A_1 = (-\infty, x_1)$  and  $A_2 = (-\infty, x_2)$  for some  $x_1$  and  $x_2$  in  $\mathbb{R}$ . Then take  $x = \min\{x_1, x_2\}$  and so

$$A_1 \cap A_2 = (-\infty, x_1) \cap (-\infty, x_2) = (-\infty, x).$$

Indeed, without loss of generality, assume that  $x_1 < x_2$ . Then notice that  $A_1 \subseteq A_2$  and so  $A_1 \cap A_2 = A_1 = (-\infty, x_1)$  where  $x_1 = \min\{x_1, x_2\} = x$ . This tells us that  $A_1 \cap A_2 \in \mathcal{T}$ .

Let  $\{A_i\}_{i \in I}$  be an arbitrary collection of sets in  $\mathcal{T}$ . We have three cases to consider; that is, either for all  $i \in I$ ,  $A_i = \emptyset$  or  $A_i = \mathbb{R}$  for some  $i \in I$ . Otherwise, for all  $i \in \mathbb{N}$ ,  $A_i = (-\infty, a_i)$  where  $a_i \in \mathbb{R}$ . If  $A_i = \emptyset$  for all  $i \in I$ , then  $\bigcup_{i \in I} A_i = \emptyset$ . Since  $\emptyset \in \mathcal{T}$  by the first case, it follows that  $\bigcup_{i \in I} A_i \in \mathcal{T}$ .

If  $A_i = \mathbb{R}$  for at least one  $i \in I$ , then  $\bigcup_{i \in I} A_i = \mathbb{R}$ . Again, since  $\mathbb{R} \in \mathcal{T}$  by our first case, it follows that  $\bigcup_{i \in I} A_i \in \mathcal{T}$ .

If  $A_i = (-\infty, a_i)$  where  $a_i \in \mathbb{R}$ . Consider the set of right endpoints of each interval  $\{a_i : i \in I\}$ . Here we will consider two cases; that is, either  $\{a_i : i \in I\}$  is unbounded or bounded. If this set is unbounded, then  $a = \sup_{i \in I} a_i = \infty$ , and so

$$\bigcup_{i \in I} A_i = (-\infty, +\infty) = \mathbb{R}.$$

If this set is bounded, then  $a = \sup_{i \in I} a_i$  exists and we claim that

$$\bigcup_{i \in I} A_i = (-\infty, a).$$

Indeed, we proceed to prove this via double inclusion.

Let  $x \in \bigcup_{i \in I} A_i$ , then there exists  $j \in I$  such that  $x \in A_j = (-\infty, a_j)$  for some  $a_j \in \mathbb{R}$ . By definition of  $a$ , it follows that  $a \geq a_j$  implying that  $x \in (-\infty, a)$ . Thus, we get that  $\bigcup_{i \in I} A_i \subseteq (-\infty, a)$ .

Let  $x \in (-\infty, a)$ . Then  $x < a = \sup_{i \in I} a_i$ . Let  $\varepsilon = a - x > 0$  since  $x < a$ . By the  $\varepsilon$  characterization of the supremum, it follows that there exists an  $\ell \in I$  such that

$$a_\ell > a - \varepsilon = a - (a - x) = x.$$

This tells us that  $x < a_\ell$  and so  $x \in (-\infty, a_\ell) \subseteq \bigcup_{i \in I} (-\infty, a_i)$ . Hence, we see that  $(-\infty, a) \subseteq \bigcup_{i \in I} (-\infty, a_i) = \bigcup_{i \in I} A_i$ . Hence, we conclude that  $\bigcup_{i \in I} A_i \in \mathcal{T}$  and so the arbitrary union of sets in  $\{A_i\}_{i \in I}$  is in  $\mathcal{T}$  is contained in  $\mathcal{T}$ .

Thus, we conclude that  $\mathcal{T}$  is a topology on  $\mathbb{R}$ . ■

(ii) Prove that  $\mathcal{T} = \{O_n : n \in \mathbb{N}\} \cup \{\emptyset, \mathbb{N}\}$  where  $O_n = \{k \in \mathbb{N} : k \geq n\}$  is a topology on  $\mathbb{N}$ .

**Proof.** Clearly,  $\emptyset, \mathbb{N}$  are in  $\mathcal{T}$  by definition of  $\mathcal{T}$ .

Let  $A_1, A_2 \in \mathcal{T}$ . Let us consider a few cases. If  $A_1 = \emptyset$  and  $A_2 = \emptyset$ , then  $A_1 \cap A_2 = \emptyset \in \mathcal{T}$ . If one of the  $A_i$ 's is empty, then  $A_1 \cap A_2 = \emptyset \in \mathcal{T}$ . If  $A_1 = \mathbb{N}$  and  $A_2 = \mathbb{N}$ , then  $A_1 \cap A_2 = \mathbb{N} \in \mathcal{T}$ . Now, suppose that  $A_1 = O_n$  and  $A_2 = O_m$  for some  $m, n \in \mathbb{N}$ . Take  $N = \max\{m, n\}$  and notice that

$$O_n \cap O_m = O_N \in \mathcal{T}$$

which tells us that  $A_1 \cap A_2 \in \mathcal{T}$ .

Indeed, without loss of generality, assume that  $m > n$ . Since  $O_m \subseteq O_n$  (since for every  $k \in O_m$ , we have  $k \geq m > n$ ), we have that  $O_m \cap O_n = O_m = O_N$  where  $N = \max\{n, m\} = m$ .

Assume that  $\{A_i\}_{i \in I}$  is an arbitrary collection of sets in  $\mathcal{T}$ . We will consider three cases; that is, either  $A_i = \emptyset$  for all  $i \in I$ ,  $A_i = I$  for some  $i \in I$ , or  $A_i = O_{n_i}$  for all  $i \in I$ .

If  $A_i = \emptyset$  for all  $i \in I$ , then it follows that  $\bigcup_{i \in I} A_i = \emptyset$ .

If there exists at least one  $j \in I$  such that  $A_j = \mathbb{N}$ , then the union will be  $\mathbb{N}$  which is in  $\mathcal{T}$ .

Otherwise, suppose  $A_i$  is in  $\{O_n : n \in \mathbb{N}\}$  for all  $i \in I$ . Then by definition,  $A_i = O_{n_i}$  where  $n_i \in \mathbb{N}$ . If we consider the set of positive integers  $S = \{n_i : i \in I\} \subseteq \mathbb{N}$ , then the well-ordering property tells us that there exists a minimum for this set; namely,  $n_0 = \min S$ . Hence, we can write

$$\bigcup_{i \in I} A_i = \bigcup_{i \in I} O_{n_i} = O_{n_0} \quad (*)$$

which is in  $\mathcal{T}$ . Indeed, notice that  $O_{n_0} \subseteq \bigcup_{i \in I} O_{n_i}$  since  $O_{n_0}$  is one of the sets in the union. On the other hand, if  $k \in \bigcup_{i \in I} O_{n_i}$ , then there exists an  $\ell \in I$  such that  $k \in O_{n_\ell}$ . Thus,  $k \geq n_\ell$ . Using the fact that  $n_0 = \min S$ , it follows that  $n_\ell \geq n_0$  and so  $k \in O_{n_0}$ . Thus,

$$\bigcup_{i \in I} O_{n_i} \subseteq O_{n_0},$$

implying (\*) from above.

Thus, it follows that  $\mathcal{T}$  is a topology on  $\mathbb{N}$ . ■

**Problem 2** (Intersections and Unions of Topologies). (i) Let  $\{\mathcal{T}_\alpha\}_{\alpha \in \Lambda}$  be a family of topologies on a nonempty set  $X$ . Prove that  $\bigcap_{\alpha \in \Lambda} \mathcal{T}_\alpha$  is a topology on  $X$ .

**Proof.** Notice that  $\emptyset, X \in \mathcal{T}_\alpha$  for all  $\alpha \in \Lambda$ . Then we have  $\emptyset, X \in \bigcap_{\alpha \in \Lambda} \mathcal{T}_\alpha$ .

Let  $A_1, A_2 \in \bigcap_{\alpha \in \Lambda} \mathcal{T}_\alpha$ . Thus, for all  $\alpha \in \Lambda$ ,  $A_1, A_2 \in \mathcal{T}_\alpha$ . Since each  $\mathcal{T}_\alpha$  is a topology, we have  $A_1 \cap A_2 \in \mathcal{T}_\alpha$  for all  $\alpha \in \Lambda$ . Hence, we have  $A_1 \cap A_2 \in \bigcap_{\alpha \in \Lambda} \mathcal{T}_\alpha$ .

Now, let  $\{A_\omega\}_{\omega \in \Omega}$  be a collection of open sets in  $\bigcap_{\alpha \in \Lambda} \mathcal{T}_\alpha$ . Then for all  $\omega \in \Omega$ ,  $A_\omega \in \mathcal{T}_\alpha$  for all  $\alpha \in \Lambda$ . That is  $\{A_\omega\}_{\omega \in \Omega}$  is a collection of open sets in  $\mathcal{T}_\alpha$  for all  $\alpha \in \Lambda$ . Thus, for all  $\alpha \in \Lambda$ ,  $\bigcup_{\omega \in \Omega} A_\omega \in \mathcal{T}_\alpha$  since each  $\mathcal{T}_\alpha$  is a topology. Thus,  $\bigcup_{\omega \in \Omega} A_\omega \in \bigcap_{\alpha \in \Lambda} \mathcal{T}_\alpha$ . Hence, we conclude that  $\bigcap_{\alpha \in \Lambda} \mathcal{T}_\alpha$  must be a topology on  $X$ . ■

(ii) Is it true that the union of two topologies on  $X$  is a topology on  $X$ ? Give a proof or a counterexample, as appropriate.

**Proof.** Let  $X = \{a, b, c\}$ . Define the following topologies  $\mathcal{T}_1 = \{X, \emptyset, \{a\}\}$  and  $\mathcal{T}_2 = \{X, \emptyset, \{b\}\}$ . Then

$$\mathcal{T}_1 \cup \mathcal{T}_2 = \{X, \emptyset, \{a\}, \{b\}\}.$$

Note that the union above is not a topology because  $\{a, b\} = \{a\} \cup \{b\} \notin \mathcal{T}_1 \cup \mathcal{T}_2$ . ■

**Problem 3** (Hausdorff condition and Metrizability). A topological space  $X$  is **Hausdorff** if for all  $x, y \in X$ ,  $x \neq y$ , there exists open sets  $U$  and  $V$  such that  $x \in U$ ,  $y \in V$ , and  $U \cap V = \emptyset$ . (This means that any two distinct points are contained in two disjoint open sets).

(i) Prove that, if a topological space is metrizable, then it is Hausdorff.

**Proof.** Let  $x, y \in X$  be distinct. Since  $X$  is metrizable, there exists a metric  $d$  such that  $d$  induces the following topology on  $X$

$$\mathcal{T} = \{A \subseteq X : \forall x \in X, \exists r > 0 \text{ such that } B_r^d(x) \subseteq A\}.$$

Define  $r = \frac{1}{2} \cdot d(x, y)$ . Since  $x \neq y$ , it follows that  $d(x, y) > 0$  and so  $r > 0$ . Define  $B_r^d(x) = U$  and  $B_r^d(y) = V$ . Notice that  $x \in U$  and  $y \in V$  by definition of those balls being centered at  $x$  and  $y$ , respectively.

We claim that  $U \cap V = \emptyset$ . It suffices to show that  $U \subseteq V^c$ . Notice that

$$V^c = [B_r^d(y)]^c = \{z \in X : d(y, z) \geq r\}.$$

To this end, let  $t \in U$  be arbitrary. Our goal is to show that

$$d(t, y) \geq r.$$

By the triangle inequality, we get that

$$\begin{aligned} d(y, x) &\leq d(y, t) + d(t, x) \\ \implies d(y, t) &\geq d(y, x) - d(t, x). \end{aligned}$$

Thus, we have that

$$\begin{aligned} d(y, t) &\geq d(y, x) - d(t, x) \\ &> d(y, x) - \frac{1}{2}d(y, x) \\ &= \frac{1}{2} \cdot d(y, x) \\ &= r. \end{aligned}$$

Hence, we see that indeed  $d(y, t) \geq r$  and so we get that  $t \in V^c$  and so  $U \cap V = \emptyset$ . Therefore,  $X$  is Hausdorff. ■

- (ii) Prove that an indiscrete topological space that contains at least two points is not metrizable.

**Proof.** Suppose for contradiction that the indiscrete topological space  $\mathcal{T}$  that contains at least two points is metrizable. By definition, it follows that there exists a metric  $d$  such that it induces  $\mathcal{T}$ . Using part (i),  $X$  is Hausdorff. That is, for all  $x, y \in X$  distinct, there exists open sets  $U$  and  $V$  in  $\mathcal{T}$  such that  $x \in U$ ,  $y \in V$ , and  $U \cap V = \emptyset$ . Since  $U, V \in \mathcal{T}$ , it follows that  $U \in \{\emptyset, X\}$  and  $V \in \{\emptyset, X\}$ . This tells us that either  $U = \emptyset$  or  $U = X$  and  $V = \emptyset$  or  $V = X$ .

If  $U = \emptyset$  and  $V = \emptyset$ , then we have a contradiction since  $x \in U$  and  $y \in V$ . If  $U = \emptyset$  and  $V = X$ , then we have another contradiction since  $x \in U$ . If  $U = X$  and  $V = \emptyset$ , then we have another contradiction since  $y \in V$ . If  $U = X$  and  $V = X$ , then we have another contradiction since  $\emptyset = U \cap V = X \cap X = X$  which tells us that  $X = \emptyset$  which is a contradiction since  $X \neq \emptyset$ . Thus, it follows that  $\mathcal{T}$  cannot be metrizable. ■

- (iii) Prove that an infinite set endowed with the cofinite topology is not metrizable.

**Proof.** Suppose for contradiction that  $\mathcal{T}$  is metrizable. Then there exists a metric  $d$  that induces  $\mathcal{T}$ . By part (i), for all  $x, y \in X$  that are distinct, there exists  $U$  and  $V$  in  $\mathcal{T}$  such that  $x \in U$ ,  $y \in V$ , and  $U \cap V = \emptyset$ . Since  $U$  and  $V$  in  $\mathcal{T}$ , it follows from De-Morgan's Laws that

$$X = X \cap X = X \cap \emptyset^c = X \setminus \emptyset = X \setminus (U \cap V) = (X \setminus U) \cup (X \setminus V).$$

Thus, the union above is finite because the complements of  $U$  and  $V$  are finite. But this tells us that  $X$  is finite which contradicts the assumption that  $X$  is infinite. Hence,  $\mathcal{T}$  must not be metrizable. ■